

Clear ["Global`*"]

solution =

DSolve[{m * x''[t] + sigma * x'[t] + k * x[t] == 0, x[0] == X, x'[0] == 0}, x[t], t]

$$\left\{ \left\{ x[t] \rightarrow \left(\left(-e^{\frac{1}{2} \left(-\frac{\sigma}{m} - \frac{\sqrt{-4 k m + \sigma^2}}{m} \right) t} \sigma + e^{\frac{1}{2} \left(-\frac{\sigma}{m} + \frac{\sqrt{-4 k m + \sigma^2}}{m} \right) t} \sigma + e^{\frac{1}{2} \left(-\frac{\sigma}{m} - \frac{\sqrt{-4 k m + \sigma^2}}{m} \right) t} \sqrt{-4 k m + \sigma^2} + e^{\frac{1}{2} \left(-\frac{\sigma}{m} + \frac{\sqrt{-4 k m + \sigma^2}}{m} \right) t} \sqrt{-4 k m + \sigma^2} \right) X \right) / \left(2 \sqrt{-4 k m + \sigma^2} \right) \right\} \right\}$$

x[t_] = FullSimplify[x[t] /. solution[[1]]]

$$e^{-\frac{\sigma t}{2 m}} X \left(\cosh \left[\frac{\sqrt{-4 k m + \sigma^2} t}{2 m} \right] + \frac{\sigma \sinh \left[\frac{\sqrt{-4 k m + \sigma^2} t}{2 m} \right]}{\sqrt{-4 k m + \sigma^2}} \right)$$

v[t_] = FullSimplify[D[x[t], {t, 1}]]

$$-\frac{2 e^{-\frac{\sigma t}{2 m}} k X \sinh \left[\frac{\sqrt{-4 k m + \sigma^2} t}{2 m} \right]}{\sqrt{-4 k m + \sigma^2}}$$

a[t_] = FullSimplify[D[v[t], {t, 1}]]

$$\left(e^{-\frac{\sigma t}{2 m}} k X \left(-\sqrt{-4 k m + \sigma^2} \cosh \left[\frac{\sqrt{-4 k m + \sigma^2} t}{2 m} \right] + \sigma \sinh \left[\frac{\sqrt{-4 k m + \sigma^2} t}{2 m} \right] \right) \right) / \left(m \sqrt{-4 k m + \sigma^2} \right)$$

Limit[x[t], t -> ∞, Assumptions -> sigma^2 > 4 * k * m && sigma > 0 && k > 0 && m > 0]

0

vv = v[t] * v[t]

$$\frac{4 e^{-\frac{\sigma t}{m}} k^2 X^2 \sinh^2 \left[\frac{\sqrt{-4 k m + \sigma^2} t}{2 m} \right]}{-4 k m + \sigma^2}$$

work = -sigma * Integrate[vv, {t, 0, ∞},

Assumptions -> sigma^2 > 4 * k * m && sigma > 0 && k > 0 && m > 0]

$$-\frac{k X^2}{2}$$

```
roots = Solve[a[t] == 0, t]
```

Solve::ifun : Inverse functions are being used by Solve, so

some solutions may not be found; use Reduce for complete solution information. >>

$$\left\{ \left\{ t \rightarrow -\frac{2 m \operatorname{ArcCosh}\left[-\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}} \right\}, \left\{ t \rightarrow \frac{2 m \operatorname{ArcCosh}\left[-\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}} \right\}, \right. \\ \left. \left\{ t \rightarrow -\frac{2 m \operatorname{ArcCosh}\left[\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}} \right\}, \left\{ t \rightarrow \frac{2 m \operatorname{ArcCosh}\left[\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}} \right\} \right\}$$

```
t1 = roots[[1]][[1]]
```

$$t \rightarrow -\frac{2 m \operatorname{ArcCosh}\left[-\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}}$$

```
t2 = roots[[2]][[1]]
```

$$t \rightarrow \frac{2 m \operatorname{ArcCosh}\left[-\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}}$$

```
t3 = roots[[3]][[1]]
```

$$t \rightarrow -\frac{2 m \operatorname{ArcCosh}\left[\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}}$$

```
t4 = roots[[4]][[1]]
```

$$t \rightarrow \frac{2 m \operatorname{ArcCosh}\left[\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}}$$

```
vm1 = Limit[v[t], t1]
```

$$-\left(\left(\frac{\sigma \operatorname{ArcCosh}\left[-\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}} \sqrt{k} \left(-2 \sqrt{k} \sqrt{m}+\sigma\right) \sqrt{\frac{2 \sqrt{k} \sqrt{m}+\sigma}{-2 \sqrt{k} \sqrt{m}+\sigma}} x\right) / \left(\sqrt{m} \sqrt{-4 k m+\sigma^2}\right)\right)$$

```
vm2 = Limit[v[t], t2]
```

$$\left(\frac{\sigma \operatorname{ArcCosh}\left[-\frac{\sigma}{2 \sqrt{k} \sqrt{m}}\right]}{\sqrt{-4 k m+\sigma^2}} \sqrt{k} \left(-2 \sqrt{k} \sqrt{m}+\sigma\right) \sqrt{\frac{2 \sqrt{k} \sqrt{m}+\sigma}{-2 \sqrt{k} \sqrt{m}+\sigma}} x\right) / \left(\sqrt{m} \sqrt{-4 k m+\sigma^2}\right)$$

vm3 = Limit[v[t], t3]

$$\left(\frac{e^{\frac{\sigma \operatorname{ArcCosh}\left[\frac{\sigma}{2\sqrt{k}\sqrt{m}}\right]}{\sqrt{-4km+\sigma^2}}} \sqrt{k} \sqrt{\frac{-2\sqrt{k}\sqrt{m}+\sigma}{2\sqrt{k}\sqrt{m}+\sigma}} (2\sqrt{k}\sqrt{m}+\sigma)x}{\left(\sqrt{m}\sqrt{-4km+\sigma^2}\right)} \right) /$$

vm4 = Limit[v[t], t4]

$$- \left(\left(\frac{e^{-\frac{\sigma \operatorname{ArcCosh}\left[\frac{\sigma}{2\sqrt{k}\sqrt{m}}\right]}{\sqrt{-4km+\sigma^2}}} \sqrt{k} \sqrt{\frac{-2\sqrt{k}\sqrt{m}+\sigma}{2\sqrt{k}\sqrt{m}+\sigma}} (2\sqrt{k}\sqrt{m}+\sigma)x}{\left(\sqrt{m}\sqrt{-4km+\sigma^2}\right)} \right) / \right)$$

m = 0.2

0.2

k = 0.1

0.1

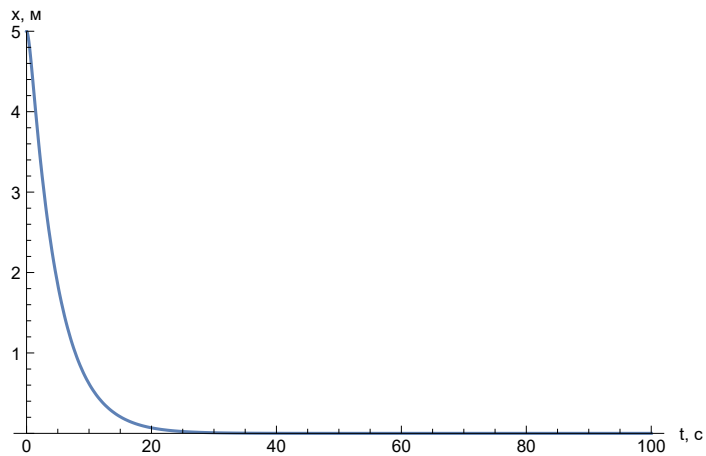
sigma = 0.5

0.5

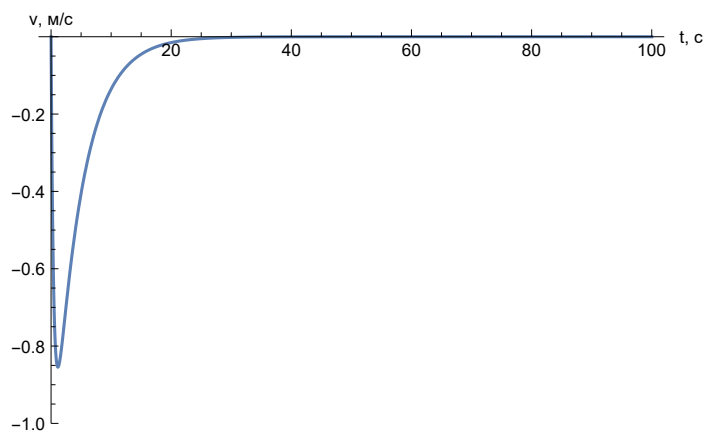
x = 5

5

Plot[{x[t]}, {t, 0, 100}, PlotRange -> {0, 5}, AxesLabel -> {"t, c", "x, m"}]



```
Plot[{v[t]}, {t, 0, 100}, PlotRange → {-1, 0}, AxesLabel → {"t, c", "v, m/c"}]
```



```
Abs[vm1]
```

```
14.6292
```

```
Abs[vm2]
```

```
0.854456
```

```
Abs[vm3]
```

```
14.6292
```

```
Abs[vm4]
```

```
0.854456
```

```
t1
```

```
t → -1.13612 - 3.04779 i
```

```
t2
```

```
t → 1.13612 + 3.04779 i
```

```
t3
```

```
t → -1.13612
```

```
t4
```

```
t → 1.13612
```